Theory Notes for the 1D Silicon p–n Junction Solver

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1 Overview

This document describes the mathematical model and numerical procedure used in the DOLFINx-based one-dimensional p–n junction solver in the repository. The code solves a steady-state drift-diffusion–Poisson system on a one-dimensional interval, using a mixed finite-element formulation for the electrostatic potential and the two quasi-Fermi energies.

The current implementation is intentionally simple:

- the device is one-dimensional in space,
- the doping profile is a smooth p-to-n transition,
- Maxwell–Boltzmann statistics are used,
- the equations are written internally in atomic units,
- each bias point is solved by Newton iteration, using the previous bias point as the initial guess.

2 Unknowns and Domain

The spatial domain is the interval

$$x \in [0, L].$$

The unknown fields are

$$\phi(x), \qquad F_n(x), \qquad F_p(x),$$

where

- $\phi(x)$ is the electrostatic potential,
- $F_n(x)$ is the electron quasi-Fermi energy,
- $F_p(x)$ is the hole quasi-Fermi energy.

The code uses atomic units internally, so the energy-like quantities ϕ, F_n, and F_p are represented in Hartree units.

3 Unit System

3.1 Atomic units used internally

The solver uses:

$$\begin{aligned} a_0 &= 5.29177210903 \times 10^{-11} \text{ m}, \\ E_h &= 27.211386245988 \text{ eV}, \\ t_0 &= 2.4188843265857 \times 10^{-17} \text{ s}, \\ k_B &= 3.166811563 \times 10^{-6} \ E_h/\text{K}. \end{aligned}$$

At $T = 300$ K,

$$k_B T \approx 9.500434689 \times 10^{-4}\ E_h.$$

3.2 Derived unit conventions

The code represents:

- length in units of a_0,
- energy and electrostatic potential in units of E_h,
- time in units of t_0,
- number density in units of a_0^{-3}.

The plots are converted back to more readable units:

- position in microns,
- doping in m^{-3},
- applied bias in volts,
- current in amperes.

4 Carrier Statistics

The solver uses Maxwell–Boltzmann carrier densities:

$$n(\phi, F_n) = N_c \exp\left(\frac{F_n + \phi - E_{c0}}{k_B T}\right), \tag{1}$$

$$p(\phi, F_p) = N_v \exp\left(\frac{E_{v0} - \phi - F_p}{k_B T}\right). \tag{2}$$

To control overflow in the numerical implementation, the exponent arguments are clipped to the interval $[-80, 80]$ before exponentiation.

5 Doping Model

The device is a p–n junction with a smooth transition centered at $x = L/2$. The donor and acceptor profiles are

$$N_D(x) = \frac{N_D^\star}{2}\left[1 + \tanh\left(\frac{x - L/2}{w}\right)\right], \tag{3}$$

$$N_A(x) = \frac{N_A^\star}{2}\left[1 - \tanh\left(\frac{x - L/2}{w}\right)\right]. \tag{4}$$

This produces a p-type left side and an n-type right side while avoiding a discontinuous step in the finite-element interpolation.

6 Governing Equations

6.1 Poisson equation

The electrostatic equation is written as

$$-\frac{d}{dx}\left(\epsilon_r \frac{d\phi}{dx}\right) = A(p - n + N_D - N_A). \tag{5}$$

Here A is the effective device cross-sectional area used in the 1D reduction. The code keeps the densities as volumetric densities and multiplies the source term by the cross-sectional area when reducing the device to one dimension.

6.2 Continuity equations

The drift-diffusion currents are modeled by

$$J_n = \mu_n \, n \frac{dF_n}{dx}, \tag{6}$$

$$J_p = \mu_p \, p \frac{dF_p}{dx}. \tag{7}$$

The steady-state continuity equations are represented in weak form, but the strong-form sign structure corresponds to

$$\frac{dJ_n}{dx} = -A\,U, \tag{8}$$

$$\frac{dJ_p}{dx} = +A\,U, \tag{9}$$

where U is the recombination source term defined below.

6.3 Recombination model

The code uses the simple recombination model

$$U = \gamma(np - n_i^2). \tag{10}$$

To stay consistent with the band-edge parameters used in the code, the intrinsic density entering this term is the mass-action value

$$n_i = \sqrt{N_c N_v}\,\exp\left(\frac{E_{v0} - E_{c0}}{2k_B T}\right). \tag{11}$$

7 Variational Formulation

Let v_ϕ, v_n, and v_p be the test functions associated with ϕ, F_n, and F_p, respectively.

7.1 Poisson weak form

The weak form used in the code is

$$\int_0^L \epsilon_r \frac{d\phi}{dx}\frac{dv_\phi}{dx}\,dx - \int_0^L A\,(p - n + N_D - N_A)\,v_\phi\,dx = 0. \tag{12}$$

7.2 Electron continuity weak form

The electron residual is

$$\int_0^L \mu_n \, n \, \frac{dF_n}{dx} \frac{dv_n}{dx} \, dx + \int_0^L A \, U \, v_n \, dx = 0. \tag{13}$$

7.3 Hole continuity weak form

The hole residual is

$$-\int_0^L \mu_p \, p \, \frac{dF_p}{dx} \frac{dv_p}{dx} \, dx + \int_0^L A \, U \, v_p \, dx = 0. \tag{14}$$

7.4 Mixed nonlinear system

The total nonlinear residual is the sum of the three weak forms:

$$\mathcal{R}(\phi, F_n, F_p; v_\phi, v_n, v_p) = \mathcal{R}_\phi + \mathcal{R}_n + \mathcal{R}_p.$$

The code solves this coupled residual directly using DOLFINx and PETSc.

8 Boundary Conditions

8.1 Ohmic contact model

The contacts are modeled as ideal Ohmic Dirichlet boundaries for all three unknowns:

$$\phi(0), \; \phi(L), \; F_n(0), \; F_n(L), \; F_p(0), \; F_p(L).$$

8.2 Equilibrium contact values

At the left p-contact the code assumes approximate charge neutrality:

$$p_L \approx N_A^\star, \tag{15}$$

$$n_L \approx \frac{n_i^2}{N_A^\star}. \tag{16}$$

At the right n-contact:

$$n_R \approx N_D^\star, \tag{17}$$

$$p_R \approx \frac{n_i^2}{N_D^\star}. \tag{18}$$

The quasi-Fermi reference level is built by inverting the Maxwell–Boltzmann relations at the contact:

$$F_n = E_{c0} - \phi + k_B T \ln\left(\frac{n}{N_c}\right), \tag{19}$$

$$F_p = E_{v0} - \phi - k_B T \ln\left(\frac{p}{N_v}\right). \tag{20}$$

The code combines the electron-based and hole-based contact electrostatic potentials by averaging the two resulting expressions at each contact.

8.3 Applied bias

For an applied voltage V, the current implementation uses the convention

$$\phi_L = \phi_{L,\mathrm{eq}}, \tag{21}$$
$$\phi_R = \phi_{R,\mathrm{eq}} + V/E_h, \tag{22}$$
$$F_{n,L} = F_0, \tag{23}$$
$$F_{p,L} = F_0, \tag{24}$$
$$F_{n,R} = F_0 - V/E_h, \tag{25}$$
$$F_{p,R} = F_0 - V/E_h. \tag{26}$$

In other words, the left contact stays fixed at equilibrium, while the right contact receives the bias shift.

9 Constants Used in the Current Solver

9.1 Material and model constants

Quantity	Value in code	Meaning
ϵ_r	11.7	silicon relative permittivity
E_{c0}	$1.12/27.211386245988\ E_h$	conduction-band reference
E_{v0}	0	valence-band reference
N_c	$2.8 \times 10^{25}\ \mathrm{m}^{-3}$	conduction-band DOS
N_v	$1.04 \times 10^{25}\ \mathrm{m}^{-3}$	valence-band DOS
n_i	$1.45 \times 10^{16}\ \mathrm{m}^{-3}$	reported intrinsic density
$N_A^\star$	$1.0 \times 10^{23}\ \mathrm{m}^{-3}$	left-side doping scale
$N_D^\star$	$1.0 \times 10^{23}\ \mathrm{m}^{-3}$	right-side doping scale
γ	$1.0 \times 10^{-20}\ \mathrm{m}^3/\mathrm{s}$	recombination parameter

9.2 Geometry and mesh

Quantity	Value in code	Meaning
L	2.0×10^{-6} m	device length
A	$1.0\ \mu\mathrm{m}^2$	effective cross-sectional area
w	2.0×10^{-8} m	junction transition width
cells	200	1D interval cells

9.3 Transport scaling

The present code uses simplified mobility scaling rather than a detailed atomic-unit conversion of the SI mobilities:

$$\mu_n = 1.0, \tag{27}$$
$$\mu_p = 0.048/0.135. \tag{28}$$

This keeps the hole mobility proportional to the electron mobility, but the values should be interpreted as pragmatic numerical scales, not as a full microscopic transport calibration.

10 Bias Sweep Procedure

The applied biases used in the current code are

$$-0.5,\ -0.4,\ -0.3,\ -0.2,\ -0.1,\ 0.0,\ 0.1,\ 0.2,\ 0.3,\ 0.4,\ 0.5,\ 0.6,\ 0.7 \quad \text{volts.}$$

The actual loop is ordered by increasing magnitude of the applied bias:

$$0,\ -0.1,\ 0.1,\ -0.2,\ 0.2,\ \ldots$$

because continuation is easier when the next point is close to the previous one.

For each new bias point:

1. the previous mixed solution is reused,
2. the Dirichlet traces are shifted to the new contact values,
3. Newton iteration is applied to the new residual,
4. the resulting fields are plotted and the right-contact current is recorded.

11 Numerical Procedure

11.1 Finite-element spaces

The code uses first-order Lagrange finite elements:

$$\phi,\ F_n,\ F_p \in \mathcal{V}_h,$$

with a mixed product space

$$\mathcal{V}_h \times \mathcal{V}_h \times \mathcal{V}_h.$$

11.2 Nonlinear solver

PETSc SNES is used through DOLFINx with:

- Newton line search,
- backtracking line search,
- direct LU factorization in the linear subproblem,
- a small nonzero LU pivot shift for robustness.

The solver tolerances currently used are:

$$\begin{aligned} \texttt{rtol} &= 10^{-8}, \\ \texttt{atol} &= 10^{-10}, \\ \texttt{max\_it} &= 80. \end{aligned}$$

11.3 Initialization strategy

At the first bias point, the code first solves the equilibrium Poisson-only problem and uses that result to initialize ϕ. The quasi-Fermi energies are initialized by linear interpolation between the two contact values.

At later bias points, the full previous mixed solution is reused and shifted to match the new boundary values.

12 Outputs

The current solver writes:

- one bias plot per applied voltage,
- an I–V curve,
- a current diagnostic plot.

Each bias plot contains:

- $\phi(x)$,
- $F_n(x)$ and $F_p(x)$,
- $N_D(x)$ and $N_A(x)$.

13 Current State of the Solver

The equilibrium point is the most reliable part of the computation. The biased drift-diffusion solve is still a numerically rough baseline rather than a polished production-quality semiconductor solver.

In particular:

- the continuation structure is in place,
- the equations and weak forms are explicit,
- the code is clean enough to iterate on,
- but the biased nonlinear solve still deserves further stabilization and physical calibration.

That is acceptable for a first educational and exploratory implementation, but it should be treated as a starting point rather than a final validated device model.